Lecture Hours for Occ 4/5/6

1. MONDAY, 2-3 PM
2. TUESDAY, 12-2 PM
3. WEDNESDAY, 4-5 PM

DK IXORA

8.3 Trigonometric Integrals

Trigonometric integrals involve algebraic combinations of the six basic trigonometric functions.

1. Basic formulas e.g. $\int \operatorname{cosec}^2 x \, dx = -\cot x + C$
2. Basic substitutions e.g. $\int \tan x \, dx = \int \frac{\sin x}{\cos x} dx$ $u = \cos x$, $du = -\sin x \, dx$ $\Rightarrow = \int \frac{-du}{u} = -\ln|u| + C$ $= -\ln|\cos x| + C$
3. Integration by parts
4. Use Identities e.g. $\int \cot^2 x \, dx$ $\left[\sin^2 x + \cos^2 x = 1 \;\Rightarrow\; 1 + \cot^2 x = \operatorname{cosec}^2 x\right] = \int \operatorname{cosec}^2 x - 1 \, dx$ $= -\cot x - x + C$
5. Multiplying by a form of 1 e.g. $\int \sec x \, dx = \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx$ $= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$ $u = \sec x + \tan x$, $du = \sec x \tan x + \sec^2 x$ $= \ln|\sec x + \tan x| + C$
6. Eliminating square roots
7. Products of powers of sines and cosines $\int \sin^m x \cos^n x \, dx$
8. Integrals of powers of $\tan x$, $\sec x$, $\csc x$, and $\cot x$
9. Products of sines and cosines (multiple angles) $\int \sin nx \cos mx \, dx$

e.g. $\int_0^{\pi} \sqrt{1 - \cos x} \, dx$ $\left[\cos x = 1 - 2\sin^2 \frac{x}{2}\right]$ $\int_0^{\pi} \sqrt{1 - (1 - 2\sin^2 \frac{x}{2})} \, dx$ $= \int_0^{\pi} \sqrt{2} \left|\sin \frac{x}{2}\right| dx = \int_0^{\pi} \sqrt{2} \sin \frac{x}{2} dx$

Products of Powers of Sines and Cosines

Integrals of the form $\int \sin^m x \cos^n x \, dx$, where m and n are nonnegative integers.

Case 1: m is odd or n is odd

- **When m is odd and n is even**

  Write $\sin^m x = \sin^{m-1} x \sin x$. $\Rightarrow$ $m-1$ even & $\sin^{m-1} x = \left[\sin^2 x\right]^{\frac{m-1}{2}}$

  Use the identity of $\sin^2 x = 1 - \cos^2 x$. $= \left(1 - \cos^2 x\right)^{\frac{m-1}{2}}$

  Then substitute $u = \cos x$. $= \left(1 - u^2\right)^{\frac{m-1}{2}}$

- **When m is even and n is odd**

  Write $\cos^n x = \cos^{n-1} x \cos x$. $n-1$ even & $\cos^{n-1} x = \left(\cos^2 x\right)^{\frac{n-1}{2}}$

  Use the identity of $\cos^2 x = 1 - \sin^2 x$. $= \left(1 - \sin^2 x\right)^{\frac{n-1}{2}}$.

  Then substitute $u = \sin x$.

- **When m and n are both odd**

  Any of the above methods can be applied.

Example:

$$\int \sin^3 x \cos^2 x \, dx = \int \sin^2 x \sin x \cos^2 x \, dx$$

$u = \cos x$

$du = -\sin x \, dx$

$$= \int (1 - \cos^2 x) \cos^2 x (\sin x \, dx)$$

$$= \int (1 - u^2) u^2 (-du)$$

$$= \int (u^4 - u^2) \, du$$

$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

Example:

$$\int \cos^5 x\, dx = \int \cos^4 x (\cos x \; dx)$$

$u = \sin x$

$du = \cos x \, dx$

$$= \int (\cos^2 x)^2 (\cos x \; dx)$$

$$= \int (1 - \sin^2 x)^2 (\cos x \, dx)$$

$$= \int (1 - u^2)^2 \, du$$

$$= \int 1 - 2u^2 + u^4 \, du$$

$$= u - \frac{2}{3}u^3 + \frac{1}{5}u^5 + C$$

$$= \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C$$

Example:

$$\int \sin^7 x \cos^5 x \, dx = \int \sin^7 x (\cos^2 x)^2 \cos x \, dx$$

$u = \sin x$

$du = \cos x \, dx$

$$= \int \sin^7 x (1 - \sin^2 x)^2 (\cos x \, dx)$$

$$= \int u^7 (1 - u^2)^2 \, du$$

$$= \int u^7 (1 - 2u^2 + u^4) \, du$$

$$= \int u^7 - 2u^9 + u^{11} \, du$$

$$= \frac{1}{8}u^8 - \frac{1}{5}u^{10} + \frac{1}{12}u^{12} + C$$

$$= \frac{1}{8}\sin^8 x - \frac{1}{5}\sin^{10} x + \frac{1}{12}\sin^{12} x + C$$

If we use $u = \cos x$ instead:

$$\int \sin^7 x \cos^5 x \, dx = \int (1 - u^2)^3 u^5 (-du)$$

$$= -\int u^5 - 3u^7 + 3u^9 - u^{11} \, du$$

$$= \frac{1}{12}u^{12} - \frac{3}{10}u^{10} + \frac{3}{8}u^8 - \frac{u^6}{6} + C = \frac{\cos^{12} x}{12} - \frac{3}{10}\cos^{10} x + \frac{3}{8}\cos^8 x - \frac{\cos^6 x}{6} + C$$

They are equal

Case 2: Both m and n are even

Recall that $\cos 2x = \cos^2 x - \sin^2 x$
$= 2\cos^2 x - 1$
$= 1 - 2\sin^2 x$.

Use the following identities:

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

To reduce the integrand to one in lower powers of $\cos 2x$.

Example:

$$\int \sin^2 x \cos^4 x \, dx = \frac{1}{16}\left(x - \frac{1}{4}\sin 4x + \frac{1}{3}\sin^3 2x\right) + C$$

$$\int \sin^2 x \cos^4 x \, dx = \int \left[\tfrac{1}{2}(1-\cos 2x)\right]\left[\tfrac{1}{2}(1+\cos 2x)\right]^2 dx$$

($\cos^4 x = (\cos^2 x)^2$)

$$= \frac{1}{8}\int (1-\cos 2x)(1+\cos 2x)^2 \, dx$$

$$= \frac{1}{8}\int (1-\cos^2 2x)(1+\cos 2x) \, dx$$

$$= \frac{1}{8}\int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) \, dx$$

($\cos^2 2x = \frac{1}{2}(1+\cos 4x)$)

$$\frac{1}{8}\int(1+\cos 2x-\cos^2 2x-\cos^3 2x)\,dx$$

$$=\frac{1}{8}\left(x+\frac{\sin 2x}{2}\right)-\frac{1}{8}\int\cos^2 2x\,dx-\frac{1}{8}\int\cos^3 2x\,dx$$

$u=\sin 2x$

$du=2\cos 2x\,dx$

$$=\frac{1}{8}\left(x+\frac{\sin 2x}{2}\right)-\frac{1}{8}\int\frac{1}{2}(1+\cos 4x)\,dx-\frac{1}{8}\int \underbrace{\cos^2 2x}_{1-u^2}\left(\underbrace{\cos 2x\,dx}_{du/2}\right)$$

$$=\frac{1}{8}\left(x+\frac{\sin 2x}{2}\right)-\frac{1}{16}\left(x+\frac{\sin 4x}{4}\right)-\frac{1}{8}\int\frac{1-u^2}{2}\,du$$

$$=\frac{1}{16}x+\frac{1}{16}\sin 2x-\frac{1}{64}\sin 4x-\frac{1}{16}\left(u-\frac{u^3}{3}\right)+C$$

$$=\frac{1}{16}x+\frac{1}{16}\sin 2x-\frac{1}{64}\sin 4x-\frac{1}{16}\sin 2x+\frac{1}{48}\sin^3 2x+C$$

$$=\frac{1}{16}\left(x-\frac{1}{4}\sin 4x+\frac{1}{3}\sin^3 2x\right)+C$$

- $\int \cos^4 x \, dx = \int [\cos^2 x]^2 \, dx$

$= \int \left[\frac{1}{2}(1+\cos 2x)\right]^2 dx$

$= \frac{1}{4}\int (1 + 2\cos 2x + \cos^2 2x) \, dx$

$= \frac{1}{4}\left(x + \frac{\cancel{2}\sin 2x}{\cancel{2}}\right) + \frac{1}{4}\int \cos^2 2x \, dx$

$= \frac{1}{4}(x + \sin 2x) + \frac{1}{4}\int \frac{1}{2}(1+\cos 4x) \, dx$

$= \frac{1}{4}x + \frac{1}{4}\sin 2x + \frac{1}{8}x + \frac{1}{8}\cdot\frac{\sin 4x}{4} + C$

$= \frac{3}{8}x + \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$

Integrals of powers of $\tan x$, $\sec x$, $\csc x$, **and** $\cot x$

Use the identities

$$1 + \tan^2 x = \sec^2 x \quad \& \quad 1 + \cot^2 x = \csc^2 x$$

to integrate by substitution or by parts when necessary.

Example:

$$\int \tan^4 x\, dx = \int (\sec^2 x - 1)\tan^2 x \, dx$$

$$= \int \sec^2 x \tan^2 x \, dx - \int \tan^2 x \, dx$$

$$= \int \sec^2 x \tan^2 x \, dx - \int (\sec^2 x - 1)\, dx$$

$$= \int \sec^2 x \tan^2 x \, dx - \tan x + x$$

$u = \tan x$

$du = \sec^2 x \, dx$

$$= \int u^2 \, du - \tan x + x$$

$$= \frac{u^3}{3} - \tan x + x + C$$

$$= \frac{\tan^3 x}{3} - \tan x + x + C$$

Example:

$$\int \sec^3 x\,dx = \int \underbrace{\sec^2 x}_{\text{integrate}}\ \underbrace{\sec x\,dx}_{\text{differentiate}}$$

$u = \sec x \qquad dv = \sec^2 x\,dx$

$\Rightarrow du = \sec x \tan x\,dx \qquad v = \tan x$

$$= uv - \int v\,du = \sec x \tan x - \int (\tan^2 x \sec x)\,dx$$

$$= \sec x \tan x - \int (\sec^2 x - 1)\sec x\,dx$$

$$\int \sec^3 x\,dx = \sec x \tan x + \int \sec x\,dx - \int \sec^3 x\,dx$$

$\int \sec x\,dx = \ln|\sec x + \tan x|$

$$\Rightarrow 2\int \sec^3 x\,dx = \sec x \tan x + \ln|\sec x + \tan x| + C$$

$$\Rightarrow \int \sec^3 x\,dx = \frac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|) + C$$

Exercise:

$$\int \operatorname{cosec}^3 x\,dx$$

Products of Sines and Cosines (multiple angles)

Integrals of the forms

$$\int \sin mx \cos nx \, dx$$

$$\int \sin mx \sin nx \, dx$$

$$\int \cos mx \cos nx \, dx$$

$\cos(m-n)x = \cos mx \cos nx + \sin mx \sin nx$

$\cos(m+n)x = \cos mx \cos nx - \sin mx \sin nx$

$\cos(m-n)x - \cos(m+n)x = 2 \sin mx \sin nx$

Use identities

$$\sin mx \sin nx = \frac{1}{2}[\cos(m-n)x - \cos(m+n)x]$$

$$\sin mx \cos nx = \frac{1}{2}[\sin(m-n)x + \sin(m+n)x]$$

$$\cos mx \cos nx = \frac{1}{2}[\cos(m-n)x + \cos(m+n)x]$$

Recall:

$\sin(a+b) = \sin a \cos b + \cos a \sin b$

$\sin(a-b) = \sin a \cos b - \cos a \sin b$

$\cos(a+b) = \cos a \cos b - \sin a \sin b$

$\cos(a-b) = \cos a \cos b + \sin a \sin b$.

Example:

$$\int \sin 3x \cos 5x \, dx = \int \frac{1}{2}\left[\sin(3x-5x) + \sin(3x+5x)\right] dx$$

$$= \int \frac{1}{2}\left[\sin(-2x) + \sin(8x)\right] dx$$

(sin is odd)

$$= \frac{1}{2}\int -\sin 2x + \sin 8x \, dx$$

$$= \frac{1}{2}\left(\frac{\cos 2x}{2} - \frac{\cos 8x}{8}\right) + C = \frac{1}{4}\cos 2x - \frac{1}{16}\cos 8x + C$$

Example:

$$\int \cos 3x \cos 4x \, dx =$$

$$= \int \frac{1}{2}\left[\cos(3x-4x) + \cos(3x+4x)\right] dx$$

(cos is even)

$$= \frac{1}{2}\int \left[\cos(-x) + \cos 7x\right] dx = \frac{1}{2}\int \cos x + \cos 7x \, dx$$

$$= \frac{1}{2}\left(\sin x + \frac{\sin 7x}{7}\right) + C$$

$$= \frac{1}{2}\sin x + \frac{1}{14}\sin 7x + C$$

$$\int \sin 4x \sin 5x \, dx = \int \frac{1}{2}\left[\cos(4x-5x) - \cos(4x+5x)\right] dx$$

$$= \frac{1}{2}\int \cos(-x) - \cos(9x)\, dx$$

$$= \frac{1}{2}\int (\cos x - \cos 9x)\, dx$$

$$= \frac{1}{2}\sin x - \frac{1}{2}\cdot\frac{\sin 9x}{9} + C$$